

Migration Mortality on the Central Mediterranean Route

Evidence from Weather Variation

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Motivation and Identification Challenge

Question. How did the dangerousness for migrants crossing the Central Mediterranean evolve across policy periods?

Why it is hard. Total deaths depend on many moving pieces at once:

- *Volume*: departures volume change constantly depending on many push and pull factors (plus no departure variable)
- *Composition*: inflatable share and boat crowding change over time
- *"Smuggling Behaviour"*: smugglers decide when to launch boats depending on many unpredictable factors
- *Weather*: daily sea conditions are exogenous but volatile
- *Institutions*: SAR coverage (Mare Nostrum → NGO SAR → LCG → NGO block) shifts repeatedly across 2013–present

A before/after count comparison confounds all of these. We need an identifier that is *fixed by construction* and does not move with the policy.

Identification Strategy

Significant wave height (SWH) is a constant physical hazard. How this hazard translates into deaths across periods?

Main period specification: Pre-2017, Post-2017; multi-period analysis

Estimand: the change in the SWH–mortality gradient.

$$\Delta\beta = \beta_{\text{SWH}}^{\text{post}} - \beta_{\text{SWH}}^{\text{pre}}$$

Identified off within-month variation in weather (using month-year FE, absorbing t.v. confounders)

Primary specification:

$$n_t^{\text{deaths}} = \beta_1 \text{SWH}_t^{\text{prev5d}} + \beta_3 \text{SWH}_t^{\text{prev5d}} \times \text{Post}_t + \gamma_{m(t)} + \varepsilon_t$$

Regresses daily deaths on 5-day mean SWH and its interaction with a post-policy indicator, conditional on month-year fixed effects — so β_3 measures how the within-month weather-to-deaths slope changes after the policy, with all monthly-frequency confounders (volume, composition, institutions, season) absorbed by the FE.

Hypothesis:

- Weather: unchanged
- SAR: removed, provided "guardrails"

⇒ the route became more dangerous *per unit of exposure*.

Daily panel, 2014-01-15 to 2023-05-31 ($N = 3,424$ days).

Source	Variable	Notes
IOM MMP	Deaths & missing (n^{deaths})	Central Med., incident + split, drown+mixed
UNITED	Deaths & missing (n^{deaths})	Central Med., drown+mixed
ERA5 reanalysis	Prev 5-day SWH (metres)	Corridor-wide mean, 0.5° grid
Frontex Themis	Daily crossing attempts	Fr _x persons + LCG/TCG pushbacks
IMO GISIS	SAR zone polygons	Libya, Tunisia, Italy, Malta

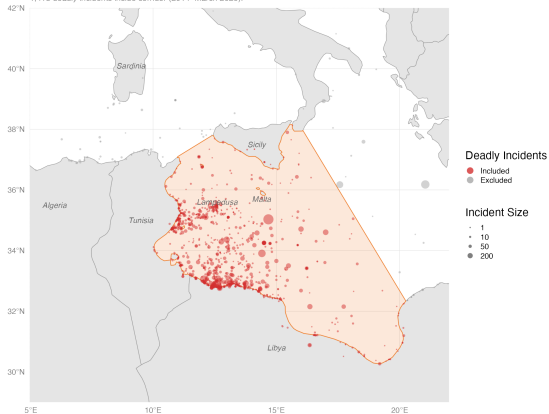
Corridor delineation. Core-corridor polygon built from ERA5 ocean cells bounded by North-Africa/EU coastlines.

SWH. spatial average over the corridor (each cell is highly correlated to the avg)

Corridor and SAR Zones

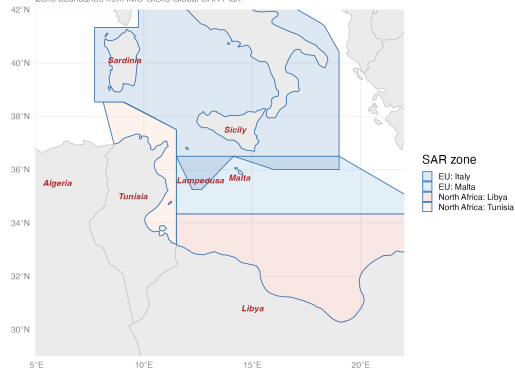
IOM Missing Migrants Project

1,413 deadly incidents inside corridor (2014–March 2026).



IMO Search and Rescue Zones

Zone boundaries from IMO GISIS Global SAR Plan.



Primary Specification — Modelling Choices

$$n_t^{\text{deaths}} \sim \text{SWH}_t^{\text{prev}} + \text{SWH}_t^{\text{prev}} \times \text{Post}_t \mid \text{month-year FE}$$

Two families:

- **Negative Binomial** (fenegbin) — high dispersion
- **Poisson QMLE** (fepois) — weaker assumptions

Agreement across the two.

Standard errors: Newey–West with lag $L = 14$ (primary); also cluster-robust on month-year.

Rate robustness: Poisson with offset $\log(\text{attempts})$ on days with non-zero attempts ($N = 2,069$).

Crossing controls: $\log 1p$ of lagged 7-day and 14-day rolling means of living crossings, included as robustness.

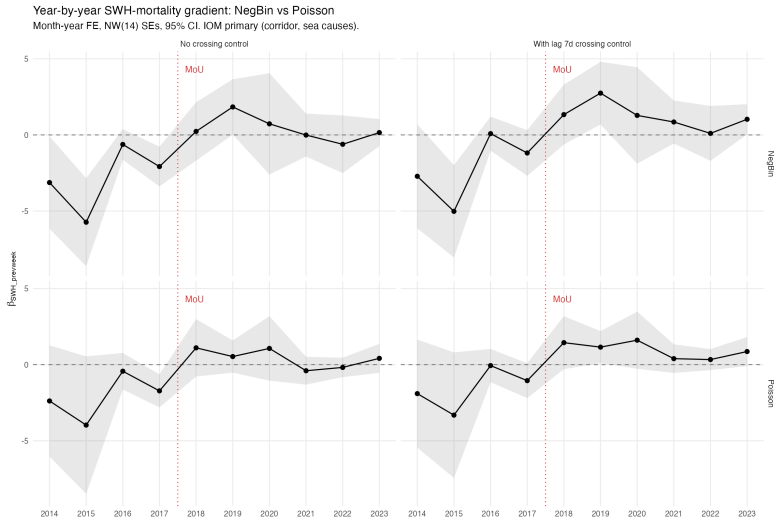
Month-year FE absorb 51% of 5-day SWH variance, leaving within-month weather shocks as the identifying variation. Post is fully absorbed by the FE $\Rightarrow$ identified only through the interaction.

Primary Results

	NegBin count	Poisson count	Poisson rate (offset)
β_1 (SWH, pre-MoU)	-1.98*** (0.60)	-1.85** (0.66)	-2.36** (0.85)
β_3 (SWH $\times$ Post)	+2.22** (0.69)	+2.04** (0.72)	+2.73** (0.93)
Observations	3,286	3,286	2,069
Month-year FE	Yes	Yes	Yes
Newey–West $L = 14$ SEs	Yes	Yes	Yes

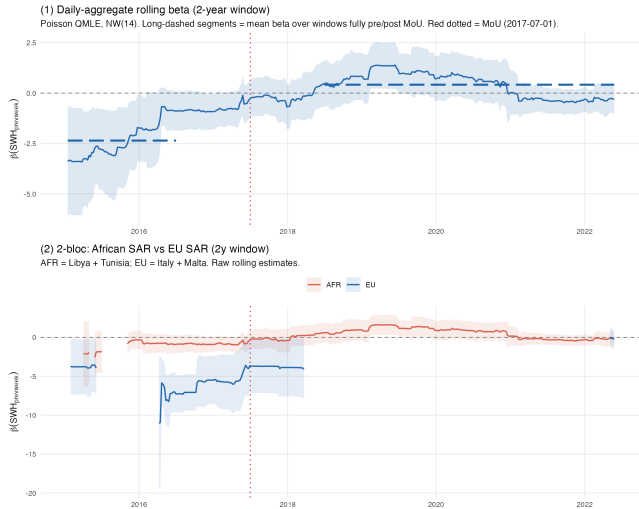
- **Sign flip:** pre-MoU $\beta(\text{SWH}) = -1.98$ (rough seas $\rightarrow$ fewer deaths, consistent with SAR protection + voluntary departure cuts). Post-MoU: $-1.98 + 2.22 = +0.24$.
- **Agreement across families** (NegBin / Poisson / rate)
- **Rate model** ($n/\text{attempts}$): the per-attempt gradient moves conditional on crossing.

Year-by-Year Gradient



$\beta(\text{SWH})$ by year. NegBin + Poisson, no crossing control and with lag-7d control. Red dotted = MoU. Gradient is negative in 2014–17, roughly zero or positive after.

Rolling-Window β — Gradual Transition



Poisson QMLE, 730-day (2-year) window, weekly step, NW(14). The shift in $\beta(\text{SWH})$ is *smooth*, not a discrete 2017-07 break — consistent with gradual SAR decline rather than a MoU policy shock.

Robustness — Zone, FE Choice, SE Variants

Zone-level (AFR vs EU SAR):

	NegBin	Poisson
β_3	+5.02*** (0.85)	+2.96** (0.93)
β_1	-4.97***	-2.69**

$\beta_3 > 0$ in both families; sign-flip preserved when exploiting within-day spatial variation across SAR blocs.

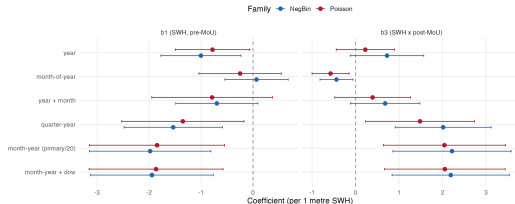
FE robustness (NegBin β_3):

- year +0.72, year+month +0.68 (insignificant)
- quarter-year +2.01***
- **month-year +2.22*** (primary)**
- FE matters: coarse FE leaves too little within-FE weather variation.

Month-year FE (primary) absorb 51% of 5-day SWH variance. Coarse FE that fail to absorb Post_t mix the interaction with a mean level shift — expected, noted in paper.

FE robustness:

FE robustness: coefficients across FE specifications
Sample: 2014-01-15 to 2023-05-31, N = 3424 days. NW(14) SEs.



Mechanism — SAR Capacity?

Replace binary Post_t **with continuous moderators** (weekly lagged averages, standardised):

- SAR share = Frontex SAR events / incidents
- Inflatable share = inflatable events / incidents
- Persons per boat (PPB) = persons / incidents

	SAR share	Inflatable share	PPB
Moderator main effect	+1.18***	+0.89**	+1.66***
SWH $\times$ Moderator	-0.63**	-0.32	-1.35***

Each column = one fit of deaths on SWH, a moderator M (z-scored, weekly) and their product, with month-year FE.

Main effect (β_M): average change in deaths when M is 1-SD higher. Mostly positive — but **confounded**: reverse causality.

Interaction (β_{int}): change in the SWH $\rightarrow$ deaths slope when M is 1-SD higher. Mechanism test — only related to how weather translates into deaths.

Only SAR $\times$ SWH is clearly negative (-0.63^{**}): 1-SD more SAR *flattens* the weather-to-deaths slope by 0.63 — the guardrail in the data. PPB $\times$ SWH is also negative but likely a volume-selection artefact (rough seas deter launches, so crowded boats only appear on calmer days).

Limitations and Takeaway

What the paper shows:

- 1 Pre-MoU: $\beta(\text{SWH}) < 0$ — rough seas produced *fewer* deaths (SAR coverage + voluntary departure cuts acted as a protective selection).
- 2 Post-MoU: $\beta(\text{SWH})$ flips toward zero / positive.
- 3 Transition is *gradual* (rolling window); MoU is a marker of a multi-year process, not a discrete cause.
- 4 SAR capacity is the mechanism most consistent with the pattern.

Limitations:

- Panel bounded by Frontex data (2014–2023, May). Earlier years (2009–2013) available only via UNITED; cross-source check shows same direction.
- IOM deaths are under-reported (Frontex–IOM cross-check: $\sim 57\%$ of on-water deaths unseen by Frontex event-level).
- Single corridor-wide SWH (diagnostic supports the simplification).
- Within-CMR identification only — EMR/WMR control routes rejected as too contaminated (EU–Turkey deal, route composition).
- Sharp MoU date is a modelling convenience; the rolling window shows the true change is smooth.